

Santosh Mallesan

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Executive Summary

Technology Executive and Verizon Fellow specializing in the commercialization of Applied Research (GenAI, LLM, Computer Vision). Holder of 33 patents and veteran of a \$40M startup exit, with a career defined by scaling global software engineering teams (80+) and delivering web-scale architectures for Fortune 500 environments. Expert at bridging the 'valley of death' between R&D/Applied Research and high-revenue commercial products.

Core Strengths

- **AI & Cloud Innovation:** Deep expertise in GenAI, LLMs, Computer Vision, and multimodal systems, applying these technologies to Web Scale platforms and Mobile architectures.
- **Executive Leadership:** Proven ability to build and manage global cross-functional teams of **80+** (US & India), overseeing budgets and large-scale software releases.
- **Startup Success & M&A:** Instrumental in scaling a startup from ground zero to **\$40M** in sales, culminating in a successful acquisition.
- **Intellectual Property Strategy:** Prolific inventor with **33** granted patents, adept at identifying patentable IP to create product differentiation and increase corporate value.
- **Product Lifecycle Management:** Expert in delivering products from inception to maturity (SaaS, Mobile, Cloud), bridging the gap between applied research and commercial deployment.
- **Domain Versatility:** Delivered solutions across diverse sectors including Telecom, Industry 4.0, HealthAI, Robotics, AR/VR, EdTech, and Public Safety.

Professional Experience

Associate Fellow, Applied Research | Verizon | Ashburn VA (08/19 - 12/25)

Directed strategy for Verizon's innovation portfolio, bridging Product Management and Product Engineering to transition applied research into creating minimum viable products.

- Enterprise ML Architecture: Unlocked a **\$15B** market opportunity by architecting and implementing Verizon's UAV connectivity platform; engineered a hybrid CPU/GPU backend for sub-second flight path optimization at scale, while enabling customers to meet the most recent FAA regulations on BvLOS flights.
- LLM Product Initiative: Led the engineering of a B2C RAG-LLM platform. Built a secure, scalable architecture capable of ingesting and querying private data for tens of millions of customers.
- Industry 4.0 Innovation: Spearheaded the R&D strategy for complex robotic operations in large-scale factories. Solved a previously unsolved domain challenge by designing algorithms that normalize indoor location data across diverse robot fleets, predicting path congestion and connectivity anomalies.
- Healthcare AI: Pioneered a multi-modal 'Aging-in-Place' platform integrating Brain-Computer Interfaces (BCI) and a number of robotic sensors (RGBD, IR, Acoustic), utilizing Differential Privacy and Edge-encryption to ensure HIPAA-level security. Led a team of 15, including a group of researchers from a leading university, and designed custom deep learning models for detecting activities of daily living.
- AR/VR Infrastructure: Standardized the XR Service Architecture across the enterprise, reducing application time-to-market by **40%** through reusable MEC-optimized (Mobile Edge Compute) modules.
- Computer Vision Strategy: Led a cross-functional initiative to unify Verizon's computer vision efforts. Designed an API-driven, framework-agnostic inferencing engine, successfully validating the architecture on AWS and demonstrating scalable cloud inference infrastructure.

- Public Safety Technology: Secured 2 patents for novel ML algorithms that utilize smartphone sensors to determine variable floor heights, solving a critical vertical location accuracy challenge for emergency responders. Validated the solution through demonstrations at key industry forums.
- Thought Leadership: Presented at The Web Conference 2020 (Computer Vision for IoT) and Tech Pulse 2020 (AI Inference Platform). Actively advised four startup cohorts in Verizon's 'Forward For Good' accelerator on Computer Vision, MEC and 5G product roadmaps. Won internal hackathons two years in a row.

Distinguished Engineer, Emerging Tech | Verizon | Ashburn VA (06/13 – 08/19)

Evangelized AI adoption within Verizon products, identifying key use cases for AI/ML application and created initial prototypes for the product engineering groups to build upon. Collaborating with the Verizon's Executive Briefing Centers (EBC), I presented Verizon technology advancements to key customer executives.

- Revenue & Retention AI: Designed and deployed ML models for the VZ Cloud product to predict user churn risk, directly informing retention strategies.
- Cost Reduction: Realized **\$2M** in annual OPEX savings by replacing manual network site selection with a Computer Vision-driven automation engine using satellite and street-view imagery. Pioneered unique algorithms for image stitching and object height detections.
- Smart Cities: Engineered accessibility solutions that utilize satellite and street imagery to identify parking spots and curb cuts for handicap driving route services.
- Strategic Partnerships: Secured a Statement of Work (SOW) with Target after winning an exclusive hackathon. Presented the resulting Computer Vision smart-cart solution directly to **Target's Board of Directors**.
- Next-Gen Video: Architected a Deep Learning computer vision solution to deliver real-time e-commerce experiences overlaid on linear TV and on-demand content, leading to multiple new product innovations.
- SDN Architecture: Led the evaluation of CSCF implementations for micro-services applicability, authoring the southbound interface RFI and demonstrating zero-touch provisioning for edge deployment.

Chief Architect | ZippSlip, Inc | Great Falls VA (09/11 – 06/13)

- SaaS Platform Architecture: Architected the flagship K12 communications platform using a .NET/SQL stack, replacing manual paper workflows for schools nationwide. Designed the cloud-native infrastructure on Amazon EC2, including a robust disaster recovery strategy to ensure business continuity.
- Mobile Strategy: Delivered native iOS and Android applications and a mobile-optimized web experience, significantly increasing parent engagement rates by **320%**.
- Social Network Engineering: Led an offshore team of 10 engineers in building a secure, compliant social networking engine for the school community. Features included SIS integration via SAML, granular privacy controls, and secure media sharing.

Chief Technology Officer | MobileCon LLC | Sterling VA (05/11 – 03/13)

- Startup Co-Founder: Established all cloud-based technical operations for a mobile marketing startup.
- FinTech Innovation: Engineered and deployed an SMS-based micropayment solution, handling secure transaction processing and gateway integrations, increasing micropayments by **500%**.

Senior Vice President, Product Development | ArX Mobile | Fairfax VA (10/10 – 05/11)

- Engineering Leadership: Directed software development and product operations for a SaaS-based mobile marketing platform.

- Platform Modernization: Led the engineering group in re-architecting the LAMP-based hosted solution to a SaaS model, while including deep social media integration (Facebook/Twitter APIs) and advanced analytics dashboards. This re-architecture resulted in **10x** OPEX cost reduction.
- Professional Services: Led the professional services division in delivering custom mobile web solutions and iOS applications for enterprise clients, utilizing device-detection logic for optimized content delivery.

Senior Director, Software Eng. | Genband (formerly NexTone) | Rockville MD (05/99 – 10/10)

As an early employee in this startup, rapidly grew from Senior Software Engineer to Senior Director. Scaled a global engineering organization from 0 to 80+ across the US and India, overseeing the technology integration for a successful corporate merger. Managed complete software development and delivery lifecycle, while also contributing to sales and marketing efforts.

- Global Team Management: Managed a cross-functional engineering and QA organization of **80** members across three locations (two US sites, one India). Successfully integrated a new US engineering team following a corporate merger.
- Product Delivery: Directed the release cycles for critical telecom infrastructure products including Session Border Controllers (SBC) and Realtime Session Managers (RSM). Oversaw development of products in multiple tech stacks (embedded Linux, J2EE, Big Data).
- Strategic Cost Reduction: Orchestrated the transition of major development activities to offshore teams in India, optimizing budget allocation while maintaining delivery velocity.
- Agile Transformation: Spearheaded the adoption of Agile/Scrum methodologies and continuous test automation. This process overhaul improved productivity and boosted product reliability to **99.999% (5-9s)** within 8 months. Designed engineering processes that successfully passed internal TL9000 audits.
- Reliability Engineering: Established a new engineering discipline to address architectural debt, directly achieving carrier-grade (5-9s) reliability. Led high-impact architectural changes while sustaining product development schedules for **400+** customers. Led the Systems Integration Team to redefine testing methodologies, emphasizing automated load and stress testing for high-volume customer deployments.
- Security Strategy: Chaired the Product Security Council. Conducted security audits and defined strategies for DoS mitigation and Intrusion Detection/Prevention (IDS/IDP).
- Strategic Planning: Served within the Office of the CTO to conduct technology assessments and define the company's long-term architectural direction. Founding member of the Intellectual Property Committee and leading over **15** patent filings.

Skills

- **AI/ML Strategy**: Agentic workflows, GenAI, Large Language Models (LLM), Retrieval Augmented Generation (RAG), LangChain, Computer Vision, Super Resolution, Deep Learning models, Statistical ML.
- **Cloud & Scale**: Kubernetes, Docker, AWS stack, Mobile Edge Compute (MEC), OpenShift, Minio, Kafka, Pulsar, Cassandra, MongoDB, Vector databases.
- **Software**: AI-Native Development (Vibe-coding, Cursor, GitHub Copilot), Python, Swift, Java, C++, C#, JavaScript/NodeJS, Scala, Objective-C, PyTorch, TensorFlow, Keras, MySQL, Postgres, SQL, .NET.

Education

Master of Science (M.S.), School of Electrical and Computer Engineering, Purdue University, IN
Bachelor of Engineering (B.E.), Electronics and Communications, Madurai Kamaraj University, India